CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 17: Circles

Topic-wise short notes with solved examples

This chapter is about the angles formed in a circle: at the centre, at the circumference, in a semicircle and in a cyclic quadrilateral.

1 Terms

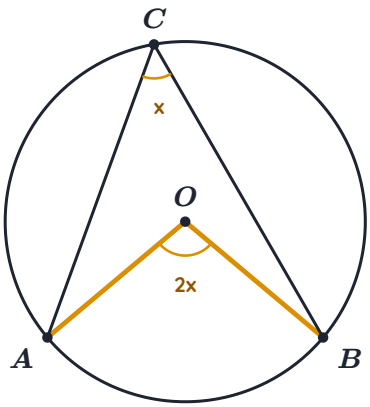
- A **chord** joins two points on a circle. A **diameter** is a chord through the centre.
- An **arc** is a part of the circle. A chord divides the circle into a **minor arc** and a **major arc**.
- A **segment** is the region between a chord and an arc. Angles "in the same segment" stand on the same chord and lie on the same side of it.
- The angle **subtended** by an arc at a point is the angle between the lines joining that point to the ends of the arc.

2 Angle at the centre and at the circumference

- **Theorem 1**: the angle that an arc subtends at the centre is **double** the angle it subtends at any point on the remaining part of the circumference.

$$\angle AOB = 2\angle ACB$$

- **Theorem 2**: angles in the same segment of a circle are **equal**.
- **Theorem 3**: the angle in a semicircle is a **right angle**. Conversely, if a chord subtends a right angle at the circumference, the chord is a diameter.



Angle AOB at the centre is double the angle ACB at the circumference

Example 2.1 O is the centre of a circle and A, B, C are points on it, with C on the major arc AB. If $\angle AOB = 110°$, find $\angle ACB$.

SOLUTION

The angle at the centre is double the angle at the circumference on the same arc.

$$\angle ACB = \frac{1}{2} \times 110°$$

$$\angle ACB = 55°$$

Example 2.2 AB is a diameter of a circle and C is a point on the circle with $\angle CAB = 35°$. Find $\angle ABC$.

SOLUTION

$\angle ACB = 90°$ (angle in a semicircle)

$$\angle ABC = 180° - 90° - 35°$$

$$\angle ABC = 55°$$

Example 2.3 O is the centre of a circle and A, B are points on it with $\angle OAB = 40°$. C is a point on the major arc. Find $\angle ACB$.

SOLUTION

$OA = OB$ (radii), so $\angle OBA = \angle OAB = 40°$

$$\angle AOB = 180° - 80° = 100°$$

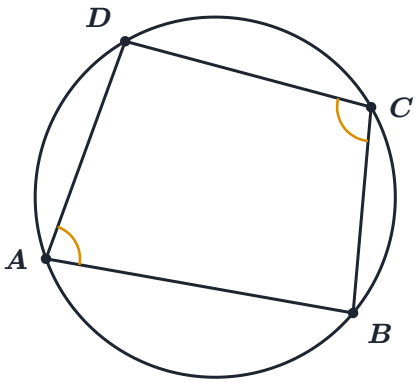
$$\angle ACB = \frac{1}{2} \times 100° = 50°$$

REMEMBER

If the point C is on the **minor** arc, use the reflex angle at the centre: $\angle ACB = \frac{1}{2}(360° - \angle AOB)$.

3 Cyclic quadrilaterals

- A quadrilateral is **cyclic** if all four of its vertices lie on a circle.
- **Theorem 4**: the opposite angles of a cyclic quadrilateral are **supplementary** (add up to $180°$).
- **Theorem 5**: an exterior angle of a cyclic quadrilateral equals the **interior opposite angle**.
- Converse: if a pair of opposite angles of a quadrilateral is supplementary, the quadrilateral is cyclic.



Cyclic quadrilateral ABCD: angle A + angle C = 180°

Example 3.1 $ABCD$ is a cyclic quadrilateral with $\angle A = 75°$ and $\angle B = 110°$. Find $\angle C$ and $\angle D$.

SOLUTION

$\angle C = 180° - \angle A = 180° - 75°$

$\angle D = 180° - \angle B = 180° - 110°$

$\angle C = 105°$ **and** $\angle D = 70°$

Example 3.2 $ABCD$ is a cyclic quadrilateral. Side AB is produced to E and $\angle CBE = 82°$. Find $\angle ADC$.

SOLUTION

The exterior angle of a cyclic quadrilateral equals the interior opposite angle.

$\angle ADC = 82°$

Example 3.3 In a cyclic quadrilateral $ABCD$, $\angle A = (2x + 4)°$ and $\angle C = (4x - 4)°$. Find x and the two angles.

SOLUTION

$(2x + 4) + (4x - 4) = 180$, so $6x = 180$ and $x = 30$

$x = 30$; $\angle A = 64°$ **and** $\angle C = 116°$

4 Arcs, chords and angles

- Equal chords of a circle (or of equal circles) subtend equal angles at the centre, and cut off equal arcs.
- Equal arcs subtend equal angles at the centre and at the circumference.
- The angle subtended by an arc is proportional to the length of the arc: an arc twice as long subtends twice the angle.
- A cyclic parallelogram is a rectangle. A cyclic trapezium is isosceles, and its diagonals are equal.

Example 4.1 A, B, C, D are points on a circle in order, with arc AB = arc BC. If $\angle ADB = 25°$, find $\angle ADC$.

SOLUTION

Equal arcs subtend equal angles at the circumference, so $\angle BDC = \angle ADB = 25°$

$\angle ADC = 50°$

5 How to write a solution

- State the property used in brackets after each step, for example "(angles in the same segment)" or "(opposite angles of a cyclic quadrilateral)".
- Look for radii: a triangle formed by two radii is isosceles.
- Look for a diameter: it gives a right angle at the circumference.
- Look for four points on the circle: they give a cyclic quadrilateral.

6 Common mistakes

- Halving the angle at the centre for a point on the wrong arc.
- Taking adjacent angles of a cyclic quadrilateral as supplementary. It is the **opposite** angles.
- Using "angles in the same segment" for angles on opposite sides of the chord.
- Giving an answer without the reason. Marks are given for the property.

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